Roll Number:_____ Name.....

Thapar Institute of Engineering & Technology, Patiala
Department of Computer Science &Engineering

B. E.(COE/CSE) : Auxiliary	**Course Code: UCS510** **Course Name: Computer Architecture and Organization**
27/Feb/2026	**5:30 PM – 8:30PM**
Time: 3 Hours, M. Marks: 100	**Faculty: SHI**

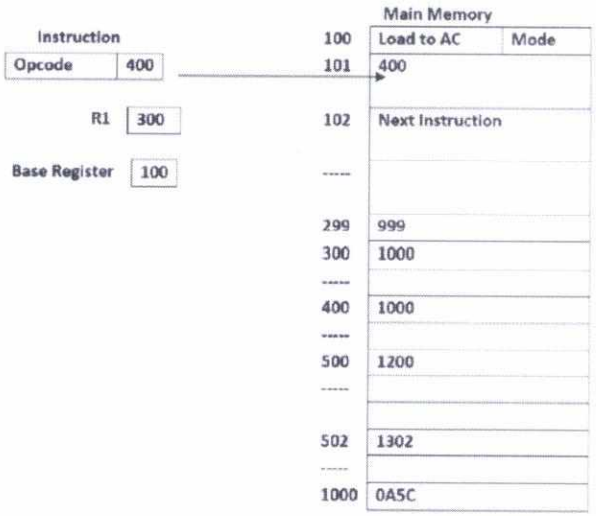
Note: All questions carry equal marks. Attempt any 5 questions in order.
Assume the missing data.

<table>
<tr><td>Q1.(A)</td><td>Design an ALU circuit with selection variable S_0 and two 2-bit data inputs A and B. The circuit generates the following four arithmetic micro-operations in conjunction with the input carry C_{in}. (Note: Draw complete circuit [Do not use any black box diagrams] for Arithmetic micro-operations given below).

<table>
<tr><td>S0</td><td>Cin</td><td>Micro-operation</td></tr>
<tr><td>0</td><td>0</td><td>A+B</td></tr>
<tr><td>0</td><td>1</td><td>A-B</td></tr>
<tr><td>1</td><td>0</td><td>Transfer A</td></tr>
<tr><td>1</td><td>1</td><td>Increment A</td></tr>
</table>
</td><td>10</td></tr>
<tr><td>Q1.(B)</td><td>Convert the following numbers into IEEE-754 single-precision (32-bit) floating point representation.

i. $(AB.75)_{16}$ ii. $(25.25)_{10}$</td><td>10</td></tr>
<tr><td>Q2. (A)</td><td>Assume 5-bit registers that hold the signed numbers. Show the step-by-step multiplication process using the Booth Algorithm where multiplicand is $(10)_{10}$ and multiplier is $(6)_{10}$.</td><td>10</td></tr>
<tr><td>Q2. (B)</td><td>What are different addressing modes, explain each of them with example. What are the advantages of using addressing mode .</td><td>10</td></tr>
<tr><td>Q3.(A)</td><td>Consider a non-pipelined processor with a clock rate of 4.2 gigahertz and an average of 6 cycles per instruction (CPI). The same processor is upgraded to a pipelined processor with 8 stages, but due to internal pipeline overhead, the clock speed is reduced to 3.0 gigahertz. Assume there is a stall delay of 0.8 ns after each stage in the pipeline. Calculate the speedup achieved by the pipelined processor compared to the non-pipelined processor. Show all calculation steps clearly.</td><td>10</td></tr>
<tr><td>Q3.(B)</td><td>What are the different Pipeline hazards, explain each of them and provide a suitable example? Write the solution to resolve each hazard.</td><td>10</td></tr>
<tr><td>Q4.(A)</td><td>A system has Main memory size of 1 MB, Cache size of 16 KB, Block size of 64 bytes and Cache is 4-way set associative. Find

1. Number of cache blocks
2. Number of sets
3. Number of bits in:
○ Block offset
○ Set index
○ Tag</td><td>10</td></tr>
</table>

Q4(B)	A system has L1 cache with Hit time = 1 ns and Hit ratio = 90%. L2 cache has Hit time = 8 ns and Hit ratio = 80%. Further the Main memory access time = 60 ns. Assume that the L2 is accessed only if L1 misses and Main memory is accessed only if L2 misses Find average memory access time.	10
Q5. A	Consider a 16-bit processor in which the main memory appears as:  Determine the effective address and operand to be loaded for the Indirect, PC relative, Base, Register Indirect, and Auto decrement using R1 addressing modes.	10
Q5 B.	a) Differentiate between memory references instructions and register reference instructions with one example of both. (4 differences only) b) Differentiate between RISC and CISC. (4 differences only) c) Differentiate between BUN and BSA instruction (2 differences only)	10
Q6 A.	A I/O device needs to transfer 10 MB of file to memory for which system opted to use DMA controller. Explain the working of DMA in step by step fashion, with all the registers, signals used. Support your answer with need and clean diagram.	10
Q6 B	A DMA controller transfers a block of 2048 bytes from an I/O device to memory. If each DMA	10